

RAJESH K. JAT

AI Engineer

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Professional Summary

AI Engineer with 2+ years of experience delivering real-time AI and analytics solutions for defense and surveillance. Skilled in building AI systems combining satellite imagery, video analytics, and edge ML. Proven expertise in CUDA-based optimization, Linux server administration, and GenAI pipeline development. Strong in designing explainable detectors and containerized AI services under strict latency constraints.

Core Competencies

- **Satellite Imagery Analytics:** Object detection (YOLOv8, Oriented Bounding Boxes), Raster data processing, geospatial feature extraction, QGIS, Rasterio.
- **Real-time Video/Pose Analytics:** Multi-object tracking, trajectory estimation, encirclement & follow pattern analysis, OpenCV, FFmpeg.
- **Edge ML & CUDA Optimization:** NVIDIA Jetson deployment, TensorRT/DeepStream integration, Linux administration, low-latency model serving.
- **GenAI/RAG Systems:** LangChain, LlamaIndex, vLLM, LLaMA, Mistral; document retrieval & summarization pipelines.
- **Backend/API Systems:** FastAPI, Docker, ReactJS, REST APIs, air-gapped deployments, performance benchmarking.

Experience

AI Research Engineer

Jul 2023 – Present

WESEE – Naval R&D Lab, New Delhi

- Developed and deployed **satellite image analytics system** using YOLOv8 with Oriented Bounding Boxes for maritime surveillance.
- Built a **real-time surveillance analytics application** capable of detecting and tracking vessels and human motion in video feeds, generating event-level metadata (timestamp, duration, type) under <50 ms latency.
- Optimized inference on **NVIDIA Jetson and CUDA-enabled Linux servers** with TensorRT quantization and tiling strategies for real-time performance.
- Designed and deployed an offline **LLM-based RAG pipeline** (LangChain + FastAPI + ReactJS) for querying classified naval documentation using vLLM inference engine.
- Integrated Coqui TTS with LLaMA for generating podcast-style document summaries, enabling multimodal situational awareness.

AI Intern

Sep 2022 – Jan 2023

Object Automation System Solutions Pvt. Ltd., Chennai

- Developed a Flask-based **collaborative filtering recommendation system** for movie suggestions.
- Hosted and administered a Linux-based Learning Management System (LMS) on a Virtual Private Server, enabling 24/7 uptime and scalable deployment.

Projects

Flying Object Flow Tracker

Built a motion analytics model to compute direction vectors and trajectory predictions for airborne objects, integrating OpenCV and NumPy-based frame-wise vector analysis.

Sponsor Analytics in Sports

Developed a YOLO-based logo detection system for analyzing sponsor placements in sports broadcasts. Computed visibility duration, screen position, and ROI metrics using real-time frame inference.

Multimodal PDF Assistant

Created a dual-modality RAG system using LLaMA 3.1 Instruct + Coqui TTS for Q&A, summarization, and text-to-speech document analytics. Powered by LangChain, ChromaDB, and Streamlit UI.

Education

M.Tech in Artificial Intelligence & Data Science

2021 – 2023

IIIT Kota, MNIT Jaipur Campus

B.Tech in Computer Science & Engineering

2017 – 2021

Rajasthan Technical University

Certifications

- AWS Fundamentals – Coursera
- Developing Cloud Native Applications – Coursera

Technical Skills

- **Languages:** Python, C++, JavaScript, SQL, R
- **Libraries/Frameworks:** PyTorch, TensorFlow, OpenCV, FastAPI, LangChain, LlamaIndex, Hugging Face, Streamlit, ReactJS, YOLO, Rasterio
- **DevOps/Tools:** Docker, Git, GitLab (Self-hosted), NVIDIA CUDA, Linux Administration
- **Databases:** MySQL, MongoDB, PostgreSQL
- **Platforms:** Ubuntu, AWS, Hadoop, RStudio